Phase B sign-off

B1-B4 are built, tested, and committed locally. B5 Volume Profile is explicitly deferred. Phase C has not started. Every capture and benchmark uses synthetic fixture data in a marked E2E database.

B1 screener / B2 replay / B3 drawing tools / B4 line alerts / B5 deferred / Phase C not started

Results and performance

Suite	Result
Backend pytest	921 passed
Mypy strict	263 checked / 0 errors
Import contracts	2 passed / 0 failed
Web unit	141 passed
Web lint / typecheck	passed / passed
Full serial E2E	227 passed / 0 failed / 0 skipped

Chart open	Samples	P50	P95	Max	Target
Warm	30	155 ms	166 ms	167 ms	<= 1,000 ms
Cold	30	482 ms	505 ms	506 ms	<= 3,000 ms

Charts	Warm open P95	Cold open P95	Pan P5	Heap max
1	168 ms	195 ms	62 fps	13 MB
4	396 ms	438 ms	61 fps	28 MB
16	1361 ms	1383 ms	61 fps	85 MB

Layout table: isolated benchmark repeat after the full suite. The 16-chart warm P95 is 1,361 ms, 5.7% above the Phase A standalone baseline of 1,288 ms (within the 10% comparison target). The full-suite sample was 1,483 ms; both raw runs are retained. This variation does not establish a cause or a measured improvement from the interval fix.
NFR-11 cold: fresh API process on port 8766 and fresh browser context for each run; navigation starts after /health/live returns 200; API boot time is excluded Hardware: Apple M4 Pro x 14, 48 GB RAM. Browser: Chromium 153.0.8010.12 (headless). Dataset: 10 synthetic symbols, 2016-08-08 to 2026-08-07, 2,610 daily bars each (marked E2E database). Grid dataset: 16 synthetic symbols, 2016-08-08 to 2026-08-07, 2,610 daily bars each (marked E2E database); MAX range. Chart-open and pan use 30 runs; heap snapshots use 6. M13 grid cold uses a new browser context while the API stays running.
The Phase A 16-chart warm P95 was 1,288 ms at 35883c9. Backend regression confirms the inactive E2ECANDLE fixture leaves live screening and rankings, RS, market breadth, and the PIT equity screen unchanged.

Workflow guide

B1. Richer screener filters and saved screens

- Open a range pill for Momentum, Buy setup, RS, Rank, or Delta Rank. Set inclusive bounds, choose a sector, and combine filters.
- Matches counts the full run. Both screeners page through every result instead of stopping at the first 100 rows.
- Save a screen as a versioned user document. It survives reload and opens from the chart bottom panel.

Proof: whole-run filter reference, saved-screen round trip, duplicate-name rejection, ETag 412 handling, validation cap, and ranking tie-break regression.

B2. EOD bar replay with no look-ahead

- Replay controls provide a date, Play/Pause, Step forward, 1x/3x/10x speed, and Exit.
- Every price and indicator series is cut at the selected date. Later corporate actions do not change earlier prices; a weekly bar contains only sessions through that date.
- Assessment uses the newest live or historical run on or before the replay date. Later drawings hide, alerts pause, and Exit restores the viewport.

Proof: bar count and date at each step, as-of requests for every series, weekly partial bar, viewport restoration, hidden drawings, and run-dated assessment and screener.

B3. Nine drawing tools in the grouped rail

- Lines: Pitchfork. Fibonacci: Trend-based extension. Shapes: Ellipse and Triangle. Annotation: Callout, Price label, and Arrow. Measure: Date range and Date and price range.
- Each tool has a create, reload, drag, resize, recolour, undo/redo, lock, and hide checklist.
- Anchors save at traded prices, so they stay attached to bars across the GOLDIAM 1:5 split.

Proof: nine per-tool checklists, split adjustment regression, hit tests, extension levels, pitchfork median and tines. The main tool captures were recaptured on E2ECANDLE; retained B1/B2/B4 workflow screenshots use synthetic screening and TCS fixtures.

B4. Alerts on a trendline, ray, or extended line

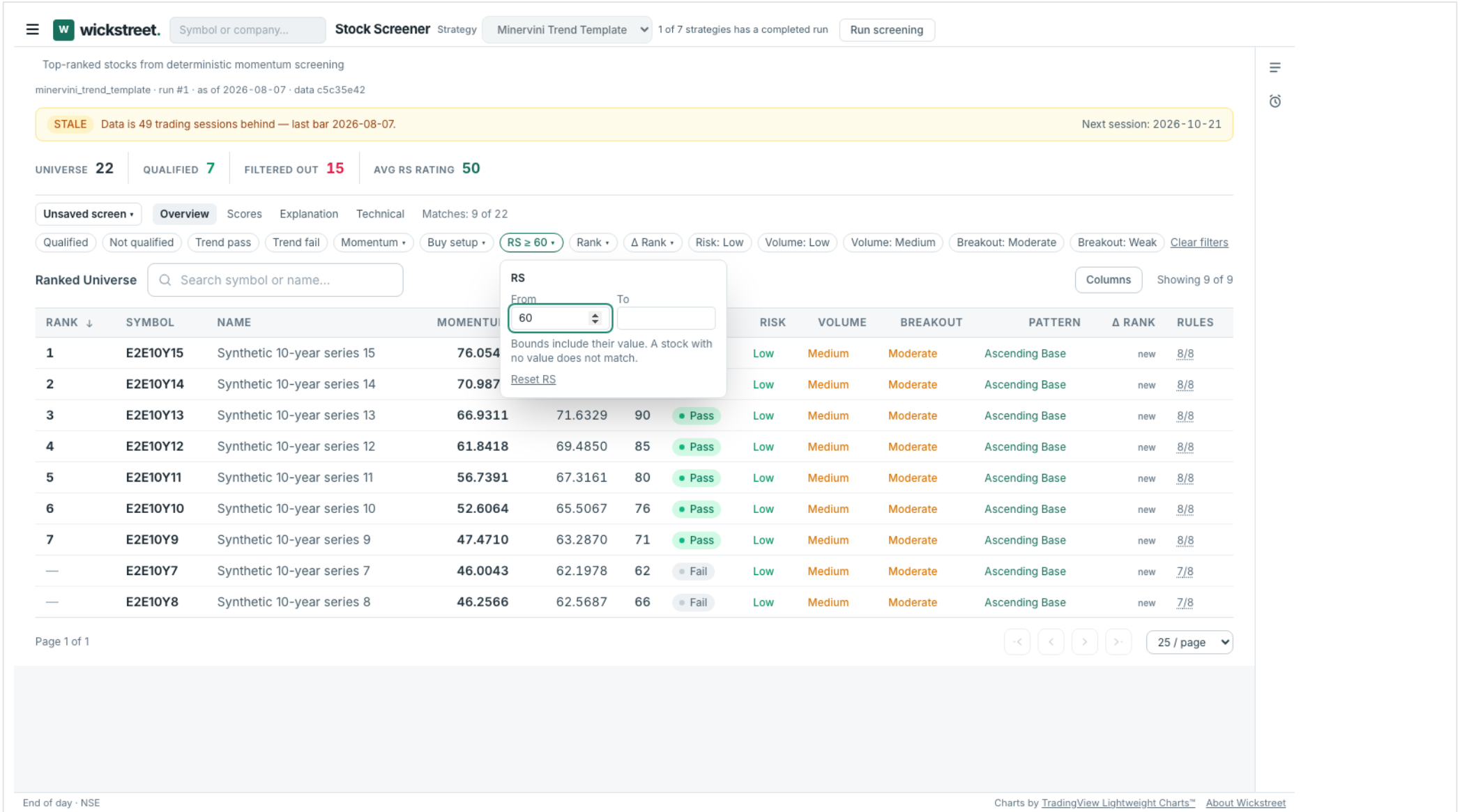
- Open an alert from a line's context menu, Shift+F10, or its Object tree entry.
- The server freezes the drawing anchors. Each anchor uses its own date's adjustment factor; each session close is compared with that date's line value.
- The first close crossing after arming fires once. Moving the drawing later does not change the saved alert rule.

Proof: alert creation, frozen snapshot after moving the drawing, oracle-predicted firing date, split and bonus handling, idempotency, stale revisions, and API rejection cases.

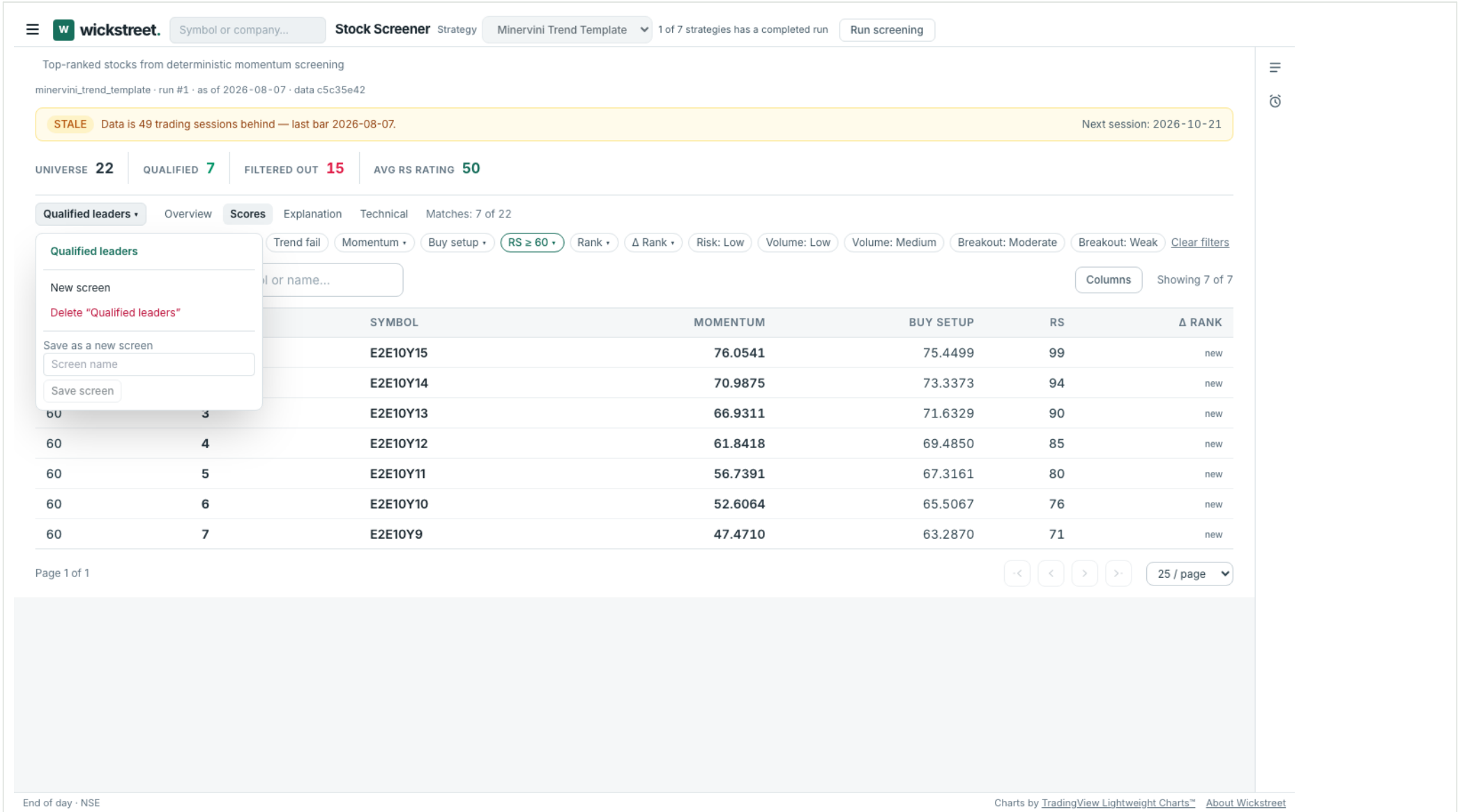
Candle review. Representative candles, intervals, themes, and replay

- E2ECANDLE is a deterministic chart-only fixture with 525 sessions from 2024-08-05 to 2026-08-07 and a 1:2 split on 2026-06-22.
- Its highs, lows, bodies, gaps, and volatility vary. The pixel test checks each drawn wick and body against the API bar within 1 px.
- The fixture is inactive and carries fund-unit ISIN INFE2ECANDL1. Backend integration coverage confirms it does not change screening results or rankings, RS ratings, market breadth, or the PIT equity screen.

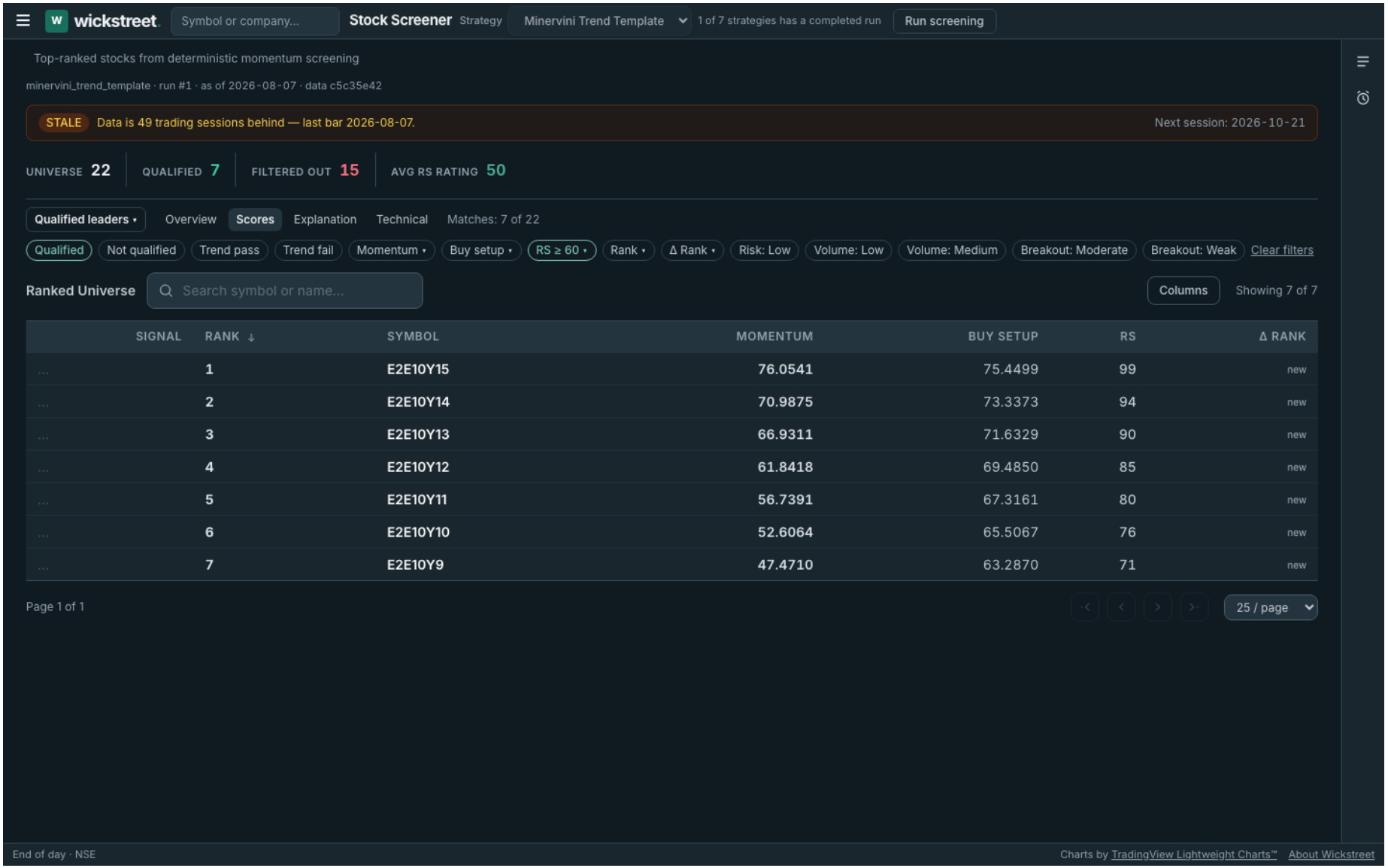
No production financial calculations changed. Axis-label collision and interval-loading viewport fixes are UI changes. The historical TCS replay and alert workflow captures remain valid synthetic evidence; these candle views were captured separately on E2ECANDLE.



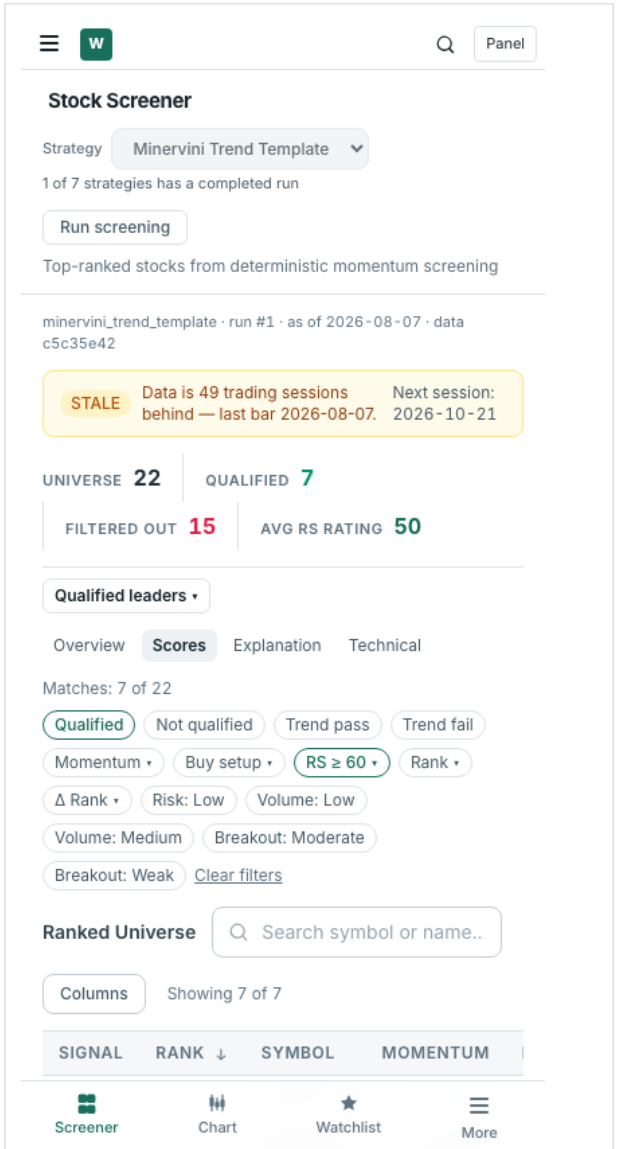
Range popover. RS >= 60 narrows the run to 9 of 22.



Saved screen menu. Qualified leaders combines Qualified, RS >= 60, and score columns.

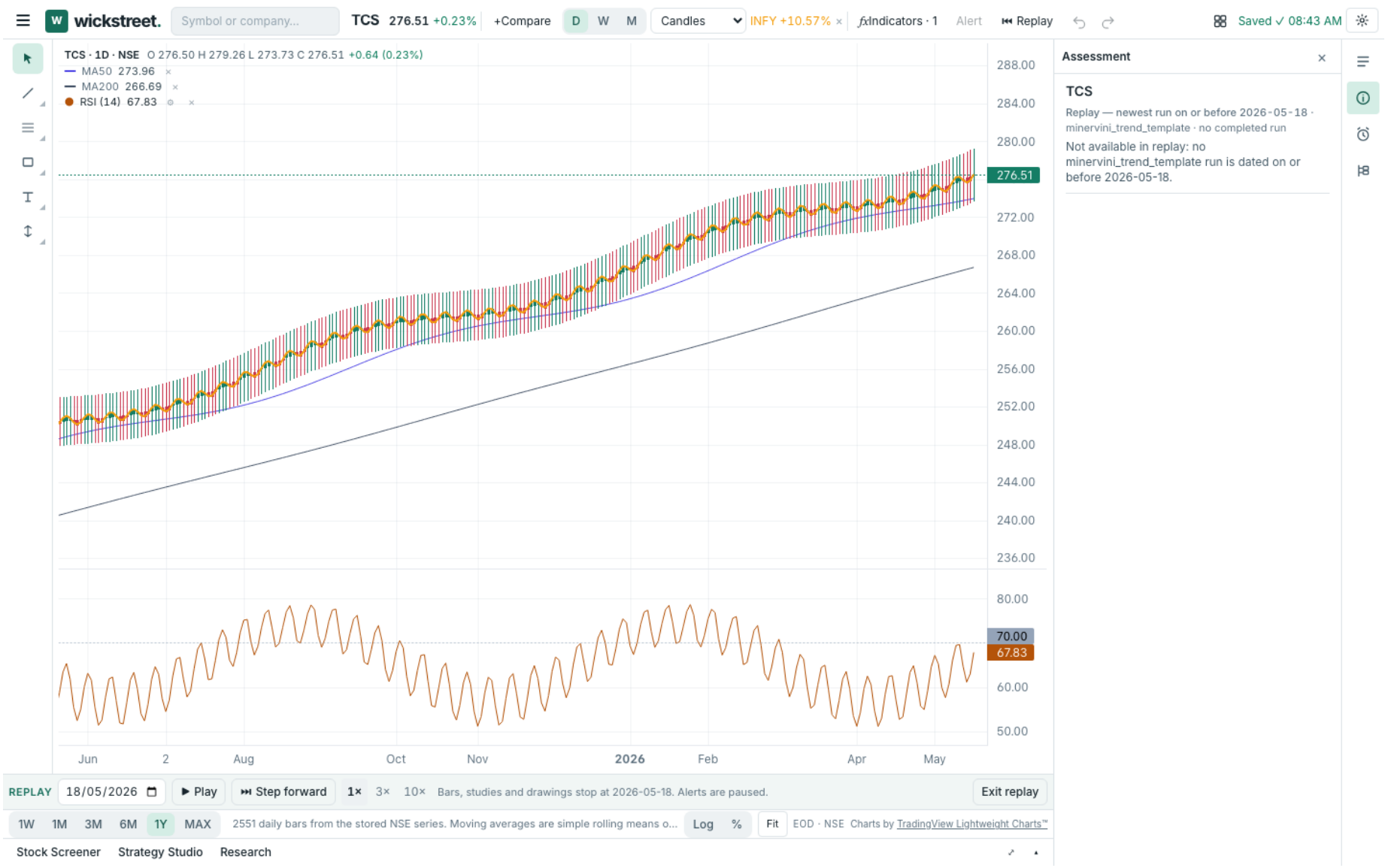


Saved screen after reload. The saved filters and columns reopen in the dark theme.



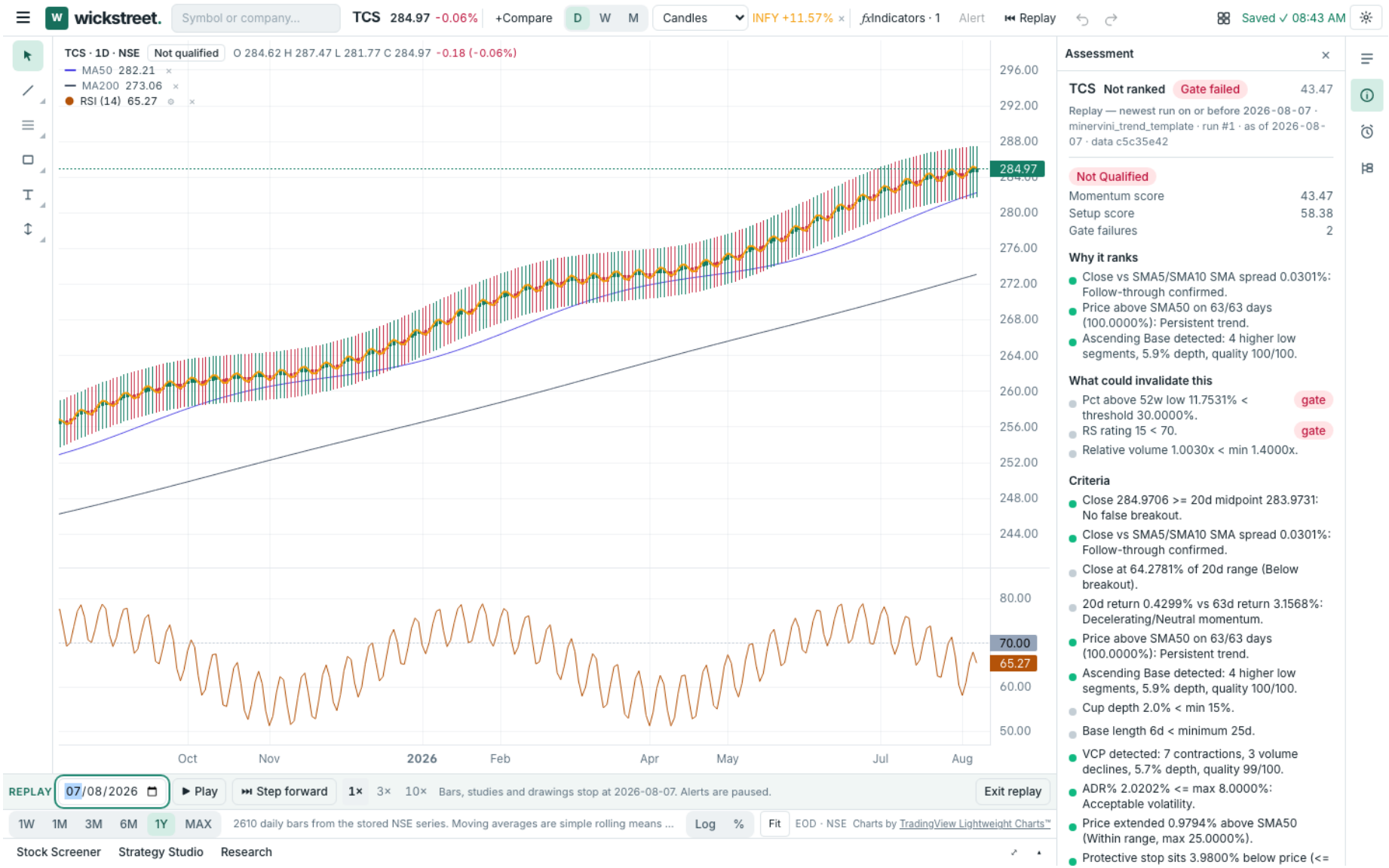
Screener at 390 px. Pills wrap without sideways scrolling.

B2 / EOD bar replay with no look-ahead

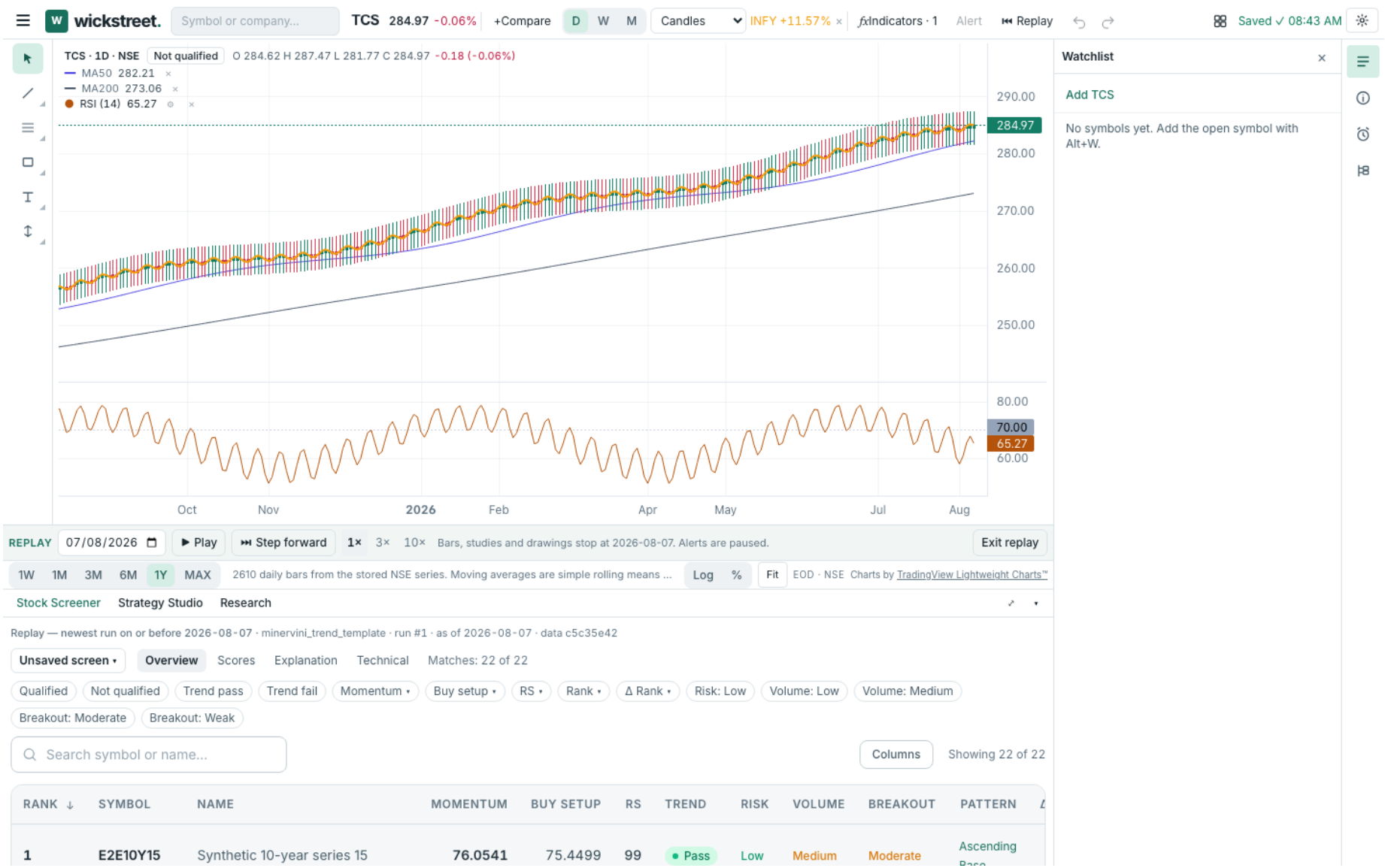


Replay before a run. Replay to 2026-05-18 shows 2,551 bars and cuts RSI and the INFY compare at that date. No run exists that early.

B2 / EOD bar replay with no look-ahead



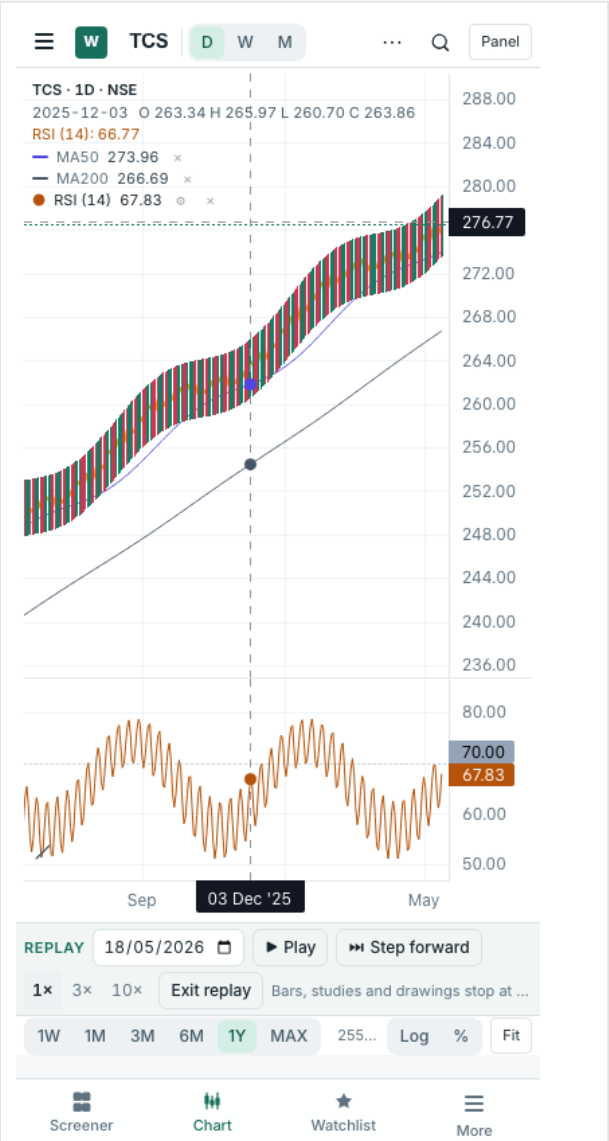
Replay on a run date. The Assessment uses the newest run on or before 2026-08-07.



Bottom screener in replay. The panel screens the same run-on-or-before date.



Replay in dark theme. Replay controls keep the sea-green accent.



Replay at 390 px. The replay row wraps while Exit stays reachable.

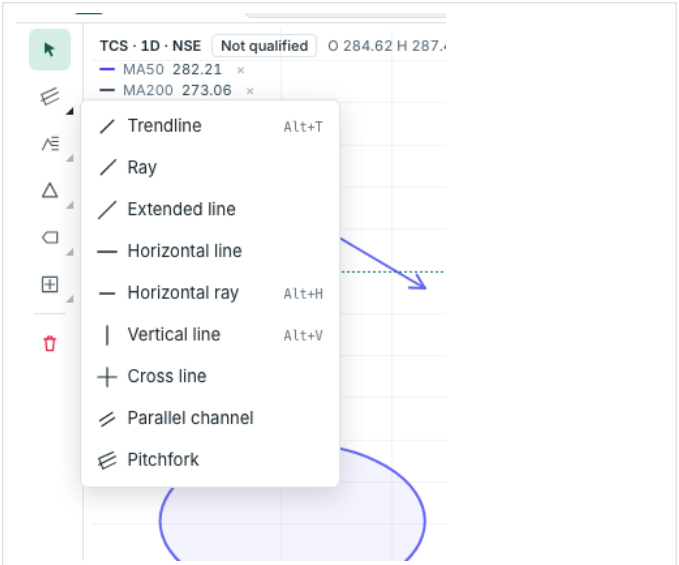


Nine tools, light theme. Final E2ECANDLE capture: Arrow, date ranges, Ellipse, Triangle, Fib extension, Pitchfork, Callout, and Price label.

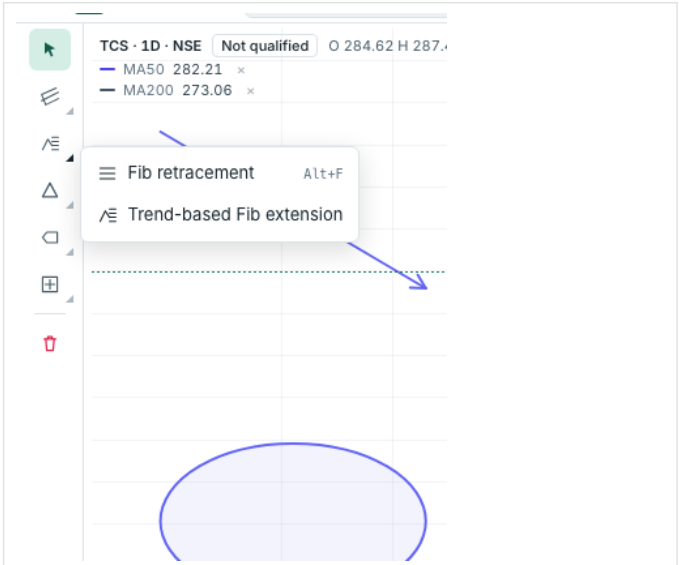


Nine tools, dark theme. The same saved drawings after reload, captured on E2ECANDLE.

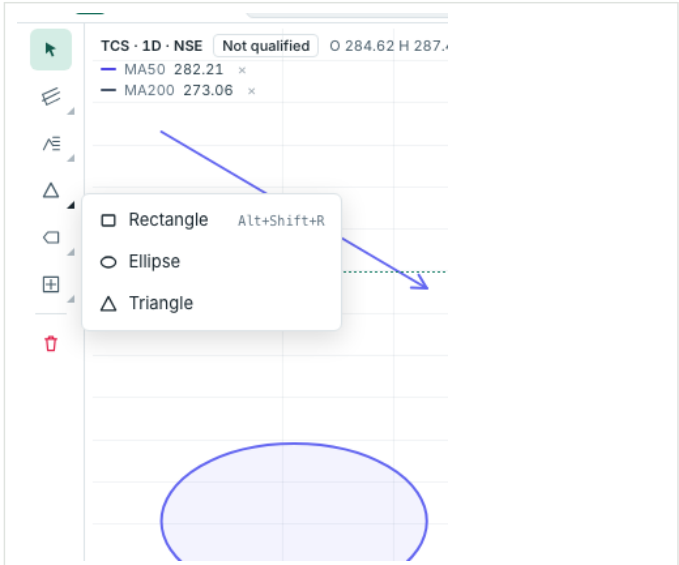
Grouped drawing tool menus



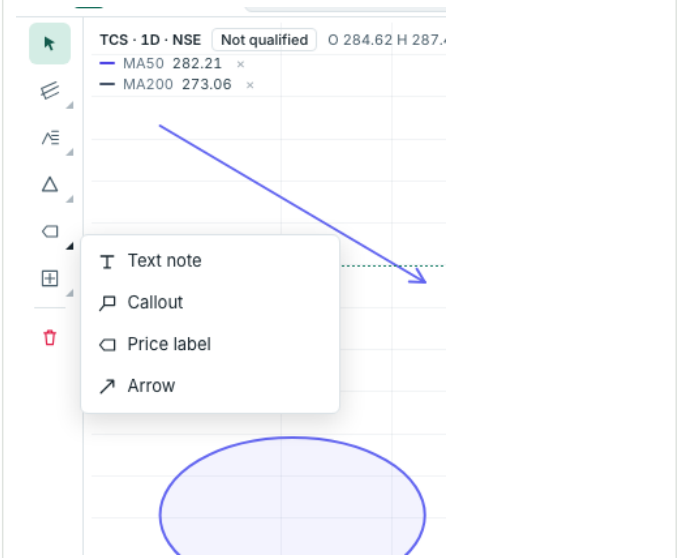
Lines. Grouped drawing tools: Lines.



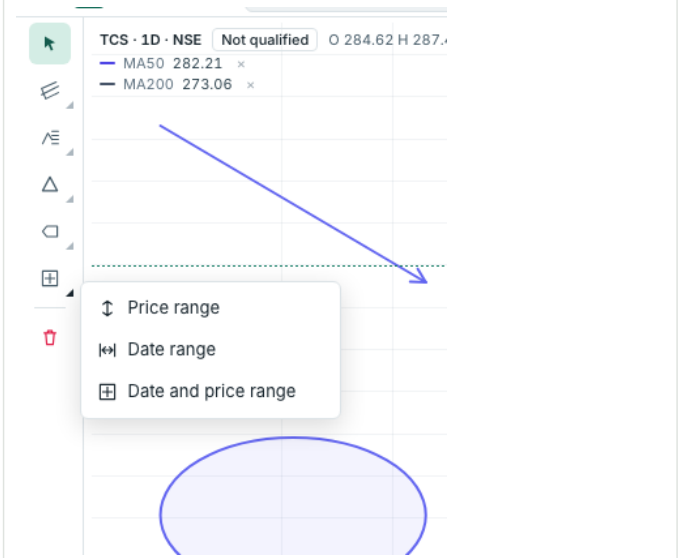
Fibonacci. Grouped drawing tools: Fibonacci.



Shapes. Grouped drawing tools: Shapes.



Annotation. Grouped drawing tools: Annotation.



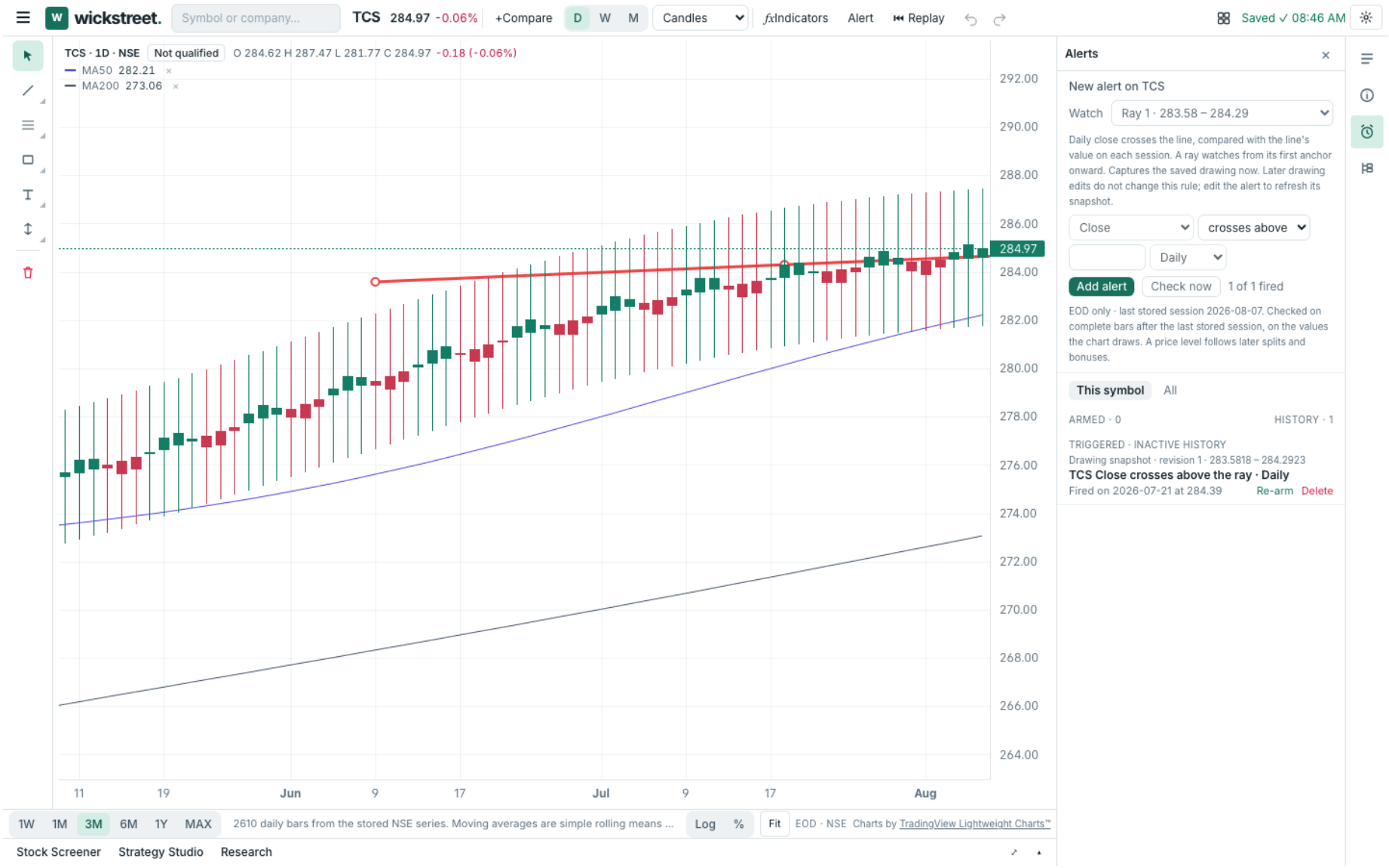
Measure. Grouped drawing tools: Measure.



Alert on this line. The drawing menu opens a line alert from the saved ray.



Prefilled ray alert. The form shows the ray and explains its frozen anchors and session rule.



Ray alert fired. The synthetic TCS ray crossing fired on 2026-07-21 at 284.39, drawing snapshot revision 1.



Daily, 6M, light. E2ECANDLE daily bars over six months.



Daily, 3M, light. E2ECANDLE daily bars over three months.



Weekly, 1Y, light. Weekly bars show the one-year range.



Monthly, MAX, light. Monthly bars show the full synthetic history.



Replay before the split. On 2026-06-12 the 1:2 split had not happened; prices remain on the pre-split basis.



Daily, 3M, dark. Daily bars in the dark theme.



Weekly, 1Y, dark. Weekly bars in the dark theme.



Phone, 3M, light. Daily chart at 390 px in the light theme.



Phone, 3M, dark. Daily chart at 390 px in the dark theme.

Decision and limits

The line-alert trigger basis is recorded in ADR-013-line-alert-amendment.md. Alerts use each anchor's date-specific adjustment factor and compare daily closes.

Known limits: replay runs on /chart and ends when the user leaves; it advances one daily session at every interval. Line alerts are close-to-close and daily. Fibonacci spacing is screen-based, so log-scale prices are not exact ratios.

Axis-label collision and interval-loading viewport fixes are UI changes. Production financial calculations are unchanged. The UI acceptance decision remains with the owner. B5 is deferred, Phase C has not started, and nothing has been pushed.

Source commit 1ed0dd8. Performance capture commit 1ed0dd8.